

Introduction to Data Management AY2026 Exercise 1 “ER Diagramming Basics”

Task 1 (up to 3 points)

You extend the sample diagram describing the basics of the university academic process. Please download this sample from Moodle (academic-er.xml). You are about including some more entities with their attributes and relationships so that to extend the given model.

Instructions

We use the browser-based **app.diagram.net** (also known as **draw.io**) diagramming tool, which works on any platform we have in computer classes.

Please access the editor at <https://app.diagrams.net/>. You can find some helpful hints at <https://github.com/jgraph/drawio/wiki/Getting-Support>. Try to experiment with the basic shapes available by using the palette toolbox. Try to put some texts inside the shapes and connect them. Save your diagram as an *xml* file and try to open the saved file. Learn how to export your diagrams in other formats, particularly such as *png*.

1. Download the sample diagram **academic-er** from Moodle.
2. Open this diagram from <https://app.diagrams.net/>.
3. Extend the diagram according to the requirements (see *Problem Description* below).
4. Save the extended diagram.
5. Export the diagram to PNG image.

DO NOT FORGET TO SAVE YOUR DIAGRAM, before leaving the browser page!

Task 1 Problem Description

The given sample diagram describes the following:

The university offers one or more programs. A program is made up of one or more courses (at least one). A student must enroll in one program. The courses are parts of the program. A program has a name attribute. A course has the following attributes: name, capacity and course ID. Students have names and IDs.

The following entities and relationships are used in the sample diagram:

- **Student** is a strong entity, with an **ID** attribute used to distinguish between students, and **name**.
- **Program** is a strong entity, with the program name.
- Each student must be enrolled in a program, so the **Student** entity is in many-to-exactly one **enrolls_in** relationship with **Program**. A program can exist without having any enrolled students.
- **Course** has meaning only in the context of a **Program**, so it's a weak entity. As a weak entity, **Course** is in many-to-one identifying relationship with its owning **Program**. It is assumed that the program must include at least one course.

Extend this diagram by supporting the following:

1. A Program has total credit points required to graduate (attribute).
2. A Teacher teaches one or more courses.
3. A Course must have at least one teacher, but may be taught by many teachers.
4. A Student has the year of first enrollment (attribute).
5. A Manager coordinates the Program and notifies students (use a simple directed association relationship).
6. There are Compulsory and Elective Courses.
7. A Program consists of Compulsory Courses.
8. A Teacher can teach several Elective Courses (but may have no Elective Courses at all).

Task 2 (up to 7 points)

For a suggested subject domain and its description, you define and describe a possible ER model containing major entities with their attributes and basic relationships between the entities. Unlike task 1, for task 2, you create **your own diagram from scratch** (no sample diagram provided).

Instructions

1. Select your problem according to individual assignments (see below how to know your problem).
2. Read the problem description (find below the Problem Description assigned to you).
3. Design your model and draw the corresponding ER diagrams.
4. Export the diagrams to a PNG image.
5. Write your explanations in a separate document. In these explanations, share your ideas of what is missing in the ER model for creating good conceptualization of the system design.

Assignment

Your individual assignment is as follows:

The remainder of division of your student ID number by 16 yields...	Your problem
0	A
1	B
2	C
3	D
4	E
5	F
6	G
7	H
8	I
9	J
10	K
11	L
12	M
13	N
14	O
15	P

Example: if the student ID is s1290045, then $\text{mod}(1290033, 16)=13$, therefore, the assigned problem is **N**.

Below you can find all the problem descriptions.

Every problem description gives you some orientation about the suggested subject domain for your ER modeling exercise. **We encourage you to be creative and to think about possible extensions of the suggested problem.**

Task 2 Problem Descriptions

Problem A. Personal Task Management

Consider designing a handy system for personalized task management that would digitalize the metaphor of the sticker board. Each sticker represents exactly one task. There are few areas on the sticker board portraying the groups of tasks according to their urgency. Also, there might be the sequences of connected tasks, where one follows another. Each task might (not necessarily) be connected to the list of responsible persons. Each task is characterized by the name (or id), its short description, its status (for example, such as scheduled, in work, suspended, completed, etc.), and the deadline. While writing the explanatory document for your solution, please make a small-scale study on the existing applications providing a functionality aiming at managing personal tasks and activities.

Problem B. Streaming Platform

A record label is interested in developing a music streaming platform that features media productions and songs from the creators. There might be different productions such as songs, video, and interactive visual works. Each production on the platform must be identified by the creator(s) and (optionally) the performer(s). Each production must be associated with at least one creator, anonymous productions are not supported. Each creator or performer is characterized by their user name (one and mandatory) and displayed names (many). It's important to note that a single production may feature multiple creators, and the role of each creator must be specified (please suggest some roles). Productions can be organized into albums. However, an album cannot exist without at least one production included in it. Users of the platform will need to sign up, provide a user ID, password, email, and select a subscription plan. The platform will offer a variety of subscription plans, each with a specific cost. Nonetheless, each user can only subscribe to one plan at a time. Users will also have the ability to create their own playlists, which can either contain different productions or remain empty.

Problem C. Hospital System

A prefecture wants to digitalize the entire healthcare system to let information be shared easily between hospitals. Each hospital will have its location, capacity, and departments. Several doctors are affiliated with a particular department, but they are restricted to working exclusively within that department. Patients have the ability to be registered at multiple hospitals, and the system's records must include their date of birth and age (age is a derived attribute). Patients can schedule appointments with specific doctors with the help of managing staff. Subsequent to the appointment, the doctor will provide a diagnosis and may prescribe medication or order the specific examinations. These prescriptions and examinations can only be authorized following an appointment. In situations where a specific test is unavailable at the patient's current hospital, they may be referred to a different hospital to undergo the required test as requested by their doctor.

Problem D. Rental Car

We are planning to create a new rental car company with multiple branches across the country. Each branch will house its own fleet of cars, sorted by categories such as sedans, vans, trucks, etc. For simplicity, cars have a flat rate per day depending on the type of car. In addition to location tracking via GPS, every car will also have a recorded history of maintenance. Customers will have the option to reserve a specific car type from a designated branch. Payments will be made on-site when the car is collected and customers must present a valid driver's license. To discourage bookings without follow-through, we will require customers to prepay for the vehicle's insurance. They will have the freedom to choose the rental date and their preferred insurance plan. We intend to provide a wide variety of insurance options.

Problem E. Food Delivery App

A start-up is planning to create a food delivery platform for countryside cities. They lack this service and sometimes food establishments are too far! Restaurants wishing to participate on this platform will be required to register their location and pickup establishment addresses. They can display their food menus along with the respective prices for each item. Users wishing to utilize the service must register, providing details such as a user ID, password, email, and address. Similarly, delivery personnel are also invited to sign up for the application. On this platform, customers have the option to order varying quantities of food from multiple restaurants in a single order. Post-purchase, customers are encouraged to leave a review for a specific restaurant from which they ordered. Likewise, delivery personnel can rate customers, which helps in identifying and possibly banning disruptive users. There are also drivers registered for specific locations. Each location is supposed to have at least one registered driver, but one driver can serve many locations.

Problem F. Performance Ticketing Service

The performing arts company launches an online ticketing service. There are a few performing halls to be registered in the system. Each performing hall is characterized by its capacity and location. Each production can be presented in at

least one venue at a time, though there are productions that can run simultaneously in the different venues. Customers can book tickets to attend many productions. Each ticket is characterized by the associated performance, seat category, performance date, floor, row, and seat number. The price is uniquely defined based on the seat category. Each production is characterized by the name, responsible stage manager(s), and the range of dates defining the period when the production is running.

Problem G. Museum Exhibition

Consider a digital service for managing a museum exhibition. The exhibition is characterized by its name, curator's name(s), display location, the latter being a room or few rooms inside the museum building. The exhibition displays a number of artifacts. Each artifact is displayed in a particular room. The artifact is characterized by its unique id, title, and creation year, as well as associated with the entity describing its creator(s). Each creator participating in the exhibition can have multiple displayed exhibits (at least one). The creation can be associated with a number of souvenir artefacts connected to this creation. There is always an exclusive label associated with a displayed artifact. Information presented on the label is to be taken from the database repository. The label can optionally be associated with the tag used by the multimedia guiding application. Each label can have at most one tag. The guiding application contains a number of articles connected with the label tags.

Problem H. Shop Selling Antique Prints

The owner of some antique shops is interested in systematization of the artifacts for sale. The shop specializes in selling antique prints. Each print is characterized by its unique id, informal name, production year, price, and the size. Some prints come with a certificate of authenticity. Each print is connected to the folder where this print is stored. There must be a folder assigned to a print. The folder can contain multiple prints, but can actually be empty. Since the folders are expected to be thematically oriented, the folder is characterized by the theme or theme(s). Suggest a few ideas of what could be the attributes of a theme. The folders are stored on the shelves. The loyal customer may have their accounts with the shop. In their accounts, they can optionally register the list of themes they are interested in. Each customer may have a list of active orders (not necessarily). The order is considered as a list of reserved prints (could not be empty) with their status data (ordered, in search, viewed, confirmed, fulfilled).

Problem I. Scientific Journal Editorial Management System

Consider the electronic system for managing scientific journal editorial activities. The journal operates with the issues. Each issue includes the articles (at least one). Each article is associated with the submission. Submission is characterized by the title, author(s), assigned editor, submission stage (for example, such as draft, initial, pending major revision, pending minor revisions, proofread, published, etc.). Each author registered in the system can have multiple submissions (but maybe no submission at all). The same concept is applied for editors. Each submission assumes a few assigned reviewers (at least one). Suggest meaningful ideas for the attributes of the entities mentioned in this description.

Problem J. Robotized Restaurant

Consider the model of a robotized restaurant. There are robots serving tables. Each table provides access to the electronic list of available dishes. The order includes the selected dishes (must have at least one). One robot's load can include dishes for different tables. A task can be assigned to each robot (such as collecting dishes from the kitchen, delivering dishes to the table, collecting the empty plates from the table, etc.). The pending tasks can be organized in the chain of requests. Each robot can access the chains of requests to know its next tasks. The restaurant's manager can access and modify the chain of requests. Each time the robot can perform only one type of task, but may be idle (no tasks).

Problem K. University Sport Club

A university wants to develop a system for managing its sports clubs. Each sports club is characterized by its clubname, sport type, short description, and activity status. Each student registered in the system is characterized by their student ID, name, email address, year of study, and contact information. A student can join one or more sports clubs, and each sports club can include many students. In each sports club, exactly one student serves as the captain, while the other students are regular members. The role of each student in a sports club must be recorded. Each sports club must have exactly one coach, and each coach is assigned to exactly one sports club. Coaches are characterized by their coach ID, name, contact information, and coaching experience. Sports clubs may use one or more university facilities for their activities, and the same facility may be shared by multiple sports clubs. Each facility is characterized by its facility name, location, capacity, and whether it is an indoor or outdoor facility. The system should also allow recording whether a facility is currently available for club activities

Problem L. Home Appliance Repair Service

A home appliance repair company wants to develop a system to manage customer repair requests. Each customer is characterized by their customer ID, name, phone number, email address, and home address. A customer can register one or more home appliances for repair, and each appliance belongs to exactly one customer. The same customer may register multiple appliances of the same type. Each appliance is identified separately by its serial number and is characterized by its appliance type, brand, and manufacture year. Common appliance types include washing machines, refrigerators, air conditioners, microwave ovens, and dishwashers. Each repair request is created for exactly one appliance, while one appliance may have multiple repair requests over time. Each repair request is characterized by its request ID, requested date, problem description, and repair status, such as pending, in progress, completed, or cancelled. Each repair request must be assigned to exactly one technician. One technician may handle zero or many repair requests. Each technician is characterized by their technician ID, name, contact information, and specialization.

Problem M. Flower Design Shop

The information system is about supporting the management activities of a flower shop. In this problem, we will try to model the aspect of communication with customers ordering flower compositions. Each customer is characterized by their customer ID, name, and contact information (phone, e-mail, list of delivery addresses). Each order is associated with a customer. The order includes one composition, greeting card, and can be characterized by the composition price and delivery price. The order can be sent using the delivery service. Each delivery can include several orders per destination but one order cannot be split between different deliveries. The order can also be collected by the customer directly from the shop (thus, there is no delivery in this case). Each composition includes 1 or more flowers and 1 or more additional decorative components (which are not necessarily flowers). The composition is also associated with a designer who created this composition. Each flower or decorative component is characterized by name, color, and price per one piece. Since not every flower can be safely combined with whatever other flower, we want that for each flower there is a list of recommendations about compatibility of this flower with other plants. The designer can access these recommendations while creating compositions.

Problem N. Travel Agency

Our travel agency is a small business specializing on individual and small-scale group tours. Each offered travel program is characterized by the main destinations (at least 1), side activities (such as visiting theaters, museums, sports events, etc.), manager of the program. The program can be implemented in the form of one or more tours. Each tour defines the means of transportation, collaborating hotels, and a list of participating customers (at least 1). Each customer is characterized by its ID, name, address, e-mail, phone number, dietary preferences. The agency also works with the professional travel guides. The guide can serve a variety of travel programs and can be assigned for a particular tour. Each guide is associated with a list of destinations (at least 1) within the scope of their professional expertise. Please think and suggest some ideas to describe the concept of destination, the ways how different destinations are connected within a program, and what are possible use cases for connecting customer interests in particular destinations.

Problem O. Orchestra Management

You are about sketching a practical model for managing a symphonic orchestra. An orchestra is a community of persons involved in different roles (conductors, musicians, administrators, librarians, etc.). In principle, each person can be involved in one or many roles. For each role, there is a number of responsibilities (please suggest what are the characteristics of a responsibility). The orchestra can participate in one or many concerts during the season. Each concert is characterized by the date, concert location, concert program. The location is characterized by its name and capacity. The concert program includes 1 or more compositions. Each composition requires a list of note scores for each instrument (actually, the orchestra librarian can be responsible for preparing the note scores for all the musicians and the conductor). There are thematically connected concerts forming groups for customer subscriptions.

Problem P. Theater Seat Reservations System

Consider a model for theater seat reservation. A theater hall consists of the groups of seats organized in two ways: by pricing (for example, S seats, A seats, B seats, etc.) and by locations (stalls, balcony, amphitheater, box, first circle, second circle, etc.). Besides this, each seat as a physical unit is characterized by its unique id, the floor, entrance door, row id, and number in a row. Instantiated seat for particular performance includes date, time, price and reservation status. Reservation status can be of three types: reserved, available for booking, pending confirmation. The customer is characterized by id, name, and e-mail. The customer can create the seat reservation order. This order is associated with the instantiated seat as one to many. Please suggest a few more entities making sense for such a system (think of your own experience of using similar systems)..

Points to note

1. Please think about the necessary attributes of each entity you decide to include in your model.
2. Please pay attention to the relationships between the entities you define.
3. Feel free to extend the suggested model with respect to your own experience. Show your creativity!

Useful Sources

In order to learn more about ER diagrams, please use the lecture materials.

In addition, for your self-study and better preparation for the exercise, the following sources are very helpful:

https://en.wikipedia.org/wiki/Entity%E2%80%93relationship_model

<https://www.smartdraw.com/entity-relationship-diagram/>

<https://creately.com/blog/diagrams/er-diagrams-tutorial/>

Grading

The grading criteria are as follows:

- Your design contains a relevant number of entities and relationships with their necessary attributes. Just for your orientation: a good solution should include 7-10 entities.
- You appropriately used strong and weak entities.
- You correctly decided on whether the relationships are one-to-one, one-to-many, zero-to-many, etc.
- Your diagram is well-structured and readable, you provided clear explanation about your models, you organized your submission as requested.

The diagram gives you up to 60% of your score, while your explanation file gives you up to 30% of your score. 10% of your score is about the carefulness of your submission preparation.

Submission

Please submit your solution as the zip package containing the following:

1. *xml* file (containing your extended version of academic-er diagram for task 1);
2. image file (*png*) – your academic-er diagram converted to a PNG image;
3. *xml* file(s) containing your ER diagram or diagrams for your solution of task 2;
4. image files (*jpg* or *png*) containing your diagram(s) converted to images;
5. document file with a brief description of your ER models for task 2. Please write some explanation of your model. You can use any document editing software. pdf is recommended for submission.